gifmakerlive Camera Technical Reference

Unified Engineering Build Spec for PR #13 + PR #14 · Repository `https://github.com/qramarq/gifmakerlive`
Prepared for Marq Williams · Document date 2026-11-05 (America/New_York) · Specs verified against manufacturer technical pages, format guides, W3C/USB-IF standards, and the open PR diffs.
Never treat this as marketing copy: numbers below are engineering inputs for `web/camera.js` (PR #13) and `web/camera-format.js` / `crates/gif-core/src/camera.rs` (PR #14).

> **Scope of this document.** One PDF covering (1) the canonical acquisition → sizing → GIF pipeline shared by both PRs; (2) major digital cinema / mirrorless / specialty UVC hardware year-by-brand with native sensor, delivery, and live HDMI/SDI/UVC feed sizes; (3) flagship smartphone camera systems 2007–2026 with browser/OS webcam modes; (4) the master resolution/aspect/constraints matrix both PRs can code against; (5) official documentation bibliography with hyperlinks.

> **Accuracy policy.** Only verified resolutions, sensor dimensions, release years, and URLs are listed. Where a year had many SKUs, the table lists the flagship / cinema-relevant model(s) that define the sensor and live-output contract browsers actually see. Gaps that PR #14 should still absorb (extra anamorphic factors 1.25× / 1.3× / 1.6×) are called out explicitly rather than invented as “already shipped.”

Table of Contents

1. Executive Architecture Spec (PR #13 ∪ PR #14)
2. Digital Cinema & Mirrorless/DSLR Brands — Technical Reference
3. Smartphones — Year-by-Year Camera Technical Reference
4. Master Resolution, Aspect Ratio & Constraints Matrix
5. Official Technical Documentation & Standards Bibliography

1. Executive Architecture Spec — Unified Build Spec for PR #13 + PR #14

1.1 How the two PRs compose one pipeline

PR #13 — Universal acquisition / transport (`web/camera.js`): enumerate every `videoinput` the browser can see (`enumerateDevices`), hot-plug via `devicechange`, open with progressive `getUserMedia` constraint fallbacks, watch live frame size via `loadedmetadata`/`resize`, negotiate `MediaRecorder` codecs (`vp9` → `vp8` → `webm` → `mp4/avc1`), and fall back to direct canvas frame-capture GIF encoding when recording is unavailable.

PR #14 — Dynamic dimension / aspect / orientation engine (`web/camera-format.js`, `crates/gif-core/src/camera.rs`, `crates/studio/src/lib.rs`): classify the live frame (smartphone vs digital-cinema), detect exact or nearest aspect profile, optionally anamorphic-desqueeze, lock portrait/landscape, switch live track modes with `applyConstraints`, and compute GIF output `width × height` under hard pixel budgets.

1. Hardware	2. Transport	3. PR #13 open	4. Live metadata	5. PR #14 size	6. GIF out
Phone sensor · Cinema camera · HDMI/SDI deck · USB-C UVC	Native MediaStream · Continuity Camera · DeviceAsWebcam ·	enumerateDevices · devicechange · constraintAttempts ·	videoWidth×videoHeight · getSettings() · getCapabilities() ·	detectCameraFormat · orientSource · outputDimensions ·	width 100–800 · MAX_FRAME_PIXELS 1.6M ·

webcam	Capture-card UVC	openCamera	resize / rVFC	placeSource	MAX_LOOP_PIXELS 80M · crop|fit

1.2 Canonical contract (hard constants both PRs already encode)

Constant	Value	Where	Meaning for implementers
`MIN_WIDTH / MAX_WIDTH`	100 / 800	PR #13 `normalizeGifWidth`; PR #14 both JS & Rust	GIF output width bounds (px)
`MAX_HEIGHT`	4096	PR #14	Hard ceiling on output height
`MAX_FRAME_PIXELS`	1_600_000	PR #14	Per-frame pixel budget
`MAX_LOOP_PIXELS / FRAME_PIXEL_BUDGET`	80_000_000	PR #13 + PR #14	Whole-GIF pixel·frame budget
`DEFAULT_CAPTURE_IDEAL`	1280×720 @ 30	PR #13	First ideal constraint attempt
`DIRECT_GIF_SECONDS`	5	PR #13	Direct frame-grab fallback duration
`MEDIA_RECORDER_LIMIT_MS`	30000	PR #13	Recorder hard stop
`DCI_ASPECT`	4096/2160 ≈ 1.8963	PR #14	DCI 17:9 exact ratio
`CAMERA_MODES`	640×480, 1280×720, 1920×1080, 3840×2160, 4096×2160, 720×1280, 1080×1920	PR #14	Live `applyConstraints` resolution switcher candidates
`RECORDER_MIME_CANDIDATES`	vp9 → vp8 → webm → mp4/avc1.42E01E → mp4/avc1 → mp4	PR #13	Codec negotiation order

1.3 Profile IDs (PR #14) — shared JS/Rust contract

profileId	Aspect / factor	Class	Notes
`match`	source aspect	auto	Default; GIF height follows live frame after orientation lock
`phone-9-16`	9/16	smartphone	Vertical phone video / browser portrait
`phone-16-9`	16/9	smartphone	Landscape phone / Continuity landscape
`phone-4-3`	4/3	smartphone	Native still sensor / Continuity 1920×1440 class
`phone-3-4`	3/4	smartphone	Portrait still / Center Stage portrait
`phone-1-1`	1	smartphone	Square crop
`dci-17-9`	4096/2160	cinema	DCI 4K / 2K
`uhd-16-9`	16/9	cinema/phone	UHD / FHD / HD delivery
`scope-239`	2.39	cinema	CinemaScope delivery
`univisium-200`	2.00	cinema	Univisium
`flat-185`	1.85	cinema	Flat theatrical
`open-gate-32`	3/2	cinema	Full-sensor Open Gate

profileId	Aspect / factor	Class	Notes
`anamorphic-43`	4/3	cinema	Squeezed 4:3 gate for 2× anamorphic
`desqueeze-133`	×1.33	desqueeze	1.33× anamorphic unsqueeze (horizontal)
`desqueeze-150`	×1.5	desqueeze	1.5×
`desqueeze-180`	×1.8	desqueeze	1.8×
`desqueeze-200`	×2.0	desqueeze	2× classic scope
`custom`	user W×H	manual	Explicit height required

> **PR reconciliation gap (intentional callout for merging #13+#14).** The user brief and common cinema glass also use **1.25×, 1.3×, and 1.6×** anamorphic squeezes (RED MONSTRO / ARRI / Hawk / Panavision families). PR #14 currently ships only 1.33 / 1.5 / 1.8 / 2.0. Recommend adding `desqueeze-125`, `desqueeze-130`, `desqueeze-160` to both JS `PROFILES` and Rust `PROFILE_IDS` before merge so the matrix and UI stay one source of truth.

1.4 Live-size & orientation rules (must not regress)

- `readLiveVideoMetadata` prefers `HTMLVideoElement.videoWidth/Height`, then `MediaStreamTrack.getSettings()`.
- `chooseLiveSize` prefers whichever of settings vs video element *just changed* (constraint switch vs rotation/resize).
- `orientSource(lock)` swaps W/H when lock is `portrait` and W>H, or `landscape` and H>W; integrates with `screen.orientation` / OS rotation.
- `outputDimensions` framing is `crop` (fill) or `fit` (letterbox inside); desqueeze forces crop and multiplies source aspect by the factor.
- `constraintAttempts` order (PR #13): exact deviceId+ideals → exact deviceId → loose deviceId → facingMode+ideals → facingMode → ideals-only → `video: true`.
- Terminal open errors (do not fall through): `NotAllowedError`, `PermissionDeniedError`, `SecurityError`.

> **Engineering takeaway for merge.** PR #13 owns the stream lifecycle; PR #14 owns every pixel decision after a frame exists. Studio wiring should call `watchLiveVideo` → `chooseLiveSize` → `detectCameraFormat` → `liveMatchSize/outputDimensions` → encoder. Do not recompute aspect inside `studio.js` ad hoc.

2. Major Digital Cinema & Mirrorless/DSLR Brands — Year-by-Year Technical Reference

Live-output column = what a browser typically receives via capture card (HDMI/SDI→UVC) or native USB-C UVC. Cinema cameras rarely expose full Open Gate over UVC; expect HD/UHD monitoring rasters unless noted.

2.1 ARRI (2005–2026)

Primary sources: ARRI ALEV Sensors tech page; ALEXA Mini LF FAQ / SUP manuals; ALEXA 35 recording-formats poster & anamorphic workflow guideline.

Year	Flagship / key model	Sensor	Physical (mm)	Native photosites / modes	Aspects	Live SDI/HDMI feed (browser-relevant)
2005–08	ARRIFLEX D-20 / D-21	Super 35 CMOS	~23.8×17.8 (D-21 era S35)	~2880×2160 class	4:3 / 16:9	HD-SDI 1920×1080
2010	ALEXA Classic	ALEV III S35	23.76×13.37 (16:9)	2880×1620 (16:9); Open Gate ~3.4K	16:9, 4:3, OG	HD/3G-SDI ≤1080p
2012–14	ALEXA Studio / Plus 4:3 / XT	ALEV III 4:3	23.76×17.82	2880×2160 (4:3); 16:9 window	4:3 ana, 16:9, 2.39	3G-SDI HD
2013–15	AMIRA	ALEV III S35	23.76×13.37	2880×1620; UHD ProRes path	16:9	3G-SDI HD / UHD on later SUP
2014	ALEXA 65 (rental)	ALEV 3 A3X	~54.1×25.6	**6560×3100**	~2.12:1 OG	3G-SDI HD monitoring
2015	ALEXA Mini	ALEV III	S35	2.8K / 3.2K ARRIRAW	16:9, 4:3	3G/6G-SDI HD–UHD
2015–17	ALEXA SXT	ALEV III	S35	2.8K–3.4K	16:9, 4:3, OG	HD-SDI / 3G
2018	ALEXA LF	ALEV 3 A2X	**36.70×25.54**	**4448×3096** LF Open Gate	OG ~1.44, LF 16:9, 2.39	6G-SDI UHD + HD
2019	ALEXA Mini LF	ALEV 3 A2X	36.70×25.54	4448×3096	same as LF	SDI1 HD; SDI2 HD or **6G UHD 3840×2160**
2022–26	ALEXA 35	ALEV 4 S35	**27.99×19.22**	**4608×3164** OG; 4608×2592 16:9; 3328×2790 6:5 (2× ana)	3:2 OG, 16:9, 6:5, 1:1	Dual **12G-SDI** HD/UHD monitoring
2024–26	ALEXA 265 (rental LF/65 family)	Large-format ALEV lineage	65-class	65mm digital gate (rental docs)	widescreen OG	SDI monitoring HD/UHD via on-set routers

Browser implication: ARRI → capture card almost always presents **1920×1080** or **3840×2160** UVC modes. Open Gate 4608×3164 is a *recording* format, not a typical UVC mode — PR #14 should detect it from uploaded clips / label hints, not expect `getUserMedia` to return 4608×3164.

2.2 RED Digital Cinema (2007–2026)

Primary sources: red.com compare-specs; MONSTRO 8K VV technical specifications; KOMODO / V-RAPTOR product pages.

Year	Model	Sensor	Size (mm) / pixels	Key modes / fps notes	Live monitor out
2007–10	RED ONE / MX	Mysterium / MX S35	~22.2×12.6; up to 4K 16:9	4K 16:9 class	HD-SDI / HDMI HD
2011–13	EPIC-X / M, SCARLET-X	Mysterium-X	S35 5K class	5K FF windows	3G-SDI / HDMI HD
2013–15	EPIC Dragon	Dragon 6K	~30.7×15.8; 6144×3160 class	6K FF, ana windows	3G-SDI HD
2015–17	WEAPON 6K / Helium 8K S35	Helium 8K S35	~29.9×15.8; 8192×4320 windowed	8K S35	3G-SDI / HDMI via expander

Year	Model	Sensor	Size (mm) / pixels	Key modes / fps notes	Live monitor out
2017–19	WEAPON / DSMC2 VV Monstro	MONSTRO 8K VV	**40.96×21.60**; **8192×4320**	8K FF 60 fps; Ana 2×/1.3×/1.25× windows documented	Expander: 1080p/720p/480p HDMI+3G-SDI
2018–20	DSMC2 Gemini / Dragon-X	Gemini 5K / Dragon-X	S35	LL / standard modes	HD monitoring
2019–21	RED Ranger	Helium/Monstro bodies	as sensor	Studio I/O	12G-SDI paths on Ranger
2021	KOMODO 6K	S35 global shutter	**27.03×14.26**; **6144×3240**	6K 17:9 40 fps; 4K 60; 2K 120	Integrated **12G-SDI** (4K60 class)
2022–24	V-RAPTOR 8K VV / XL	8K VV GS	40.96×21.60; 8192×4320	8K 120 (X); dual/triple 12G-SDI	12G-SDI 4K/HD monitoring
2023	KOMODO-X	S35 GS	27.03×14.26; 6144×3240	Faster FPS than KOMODO; USB-C; 12G-SDI	12G-SDI ≤4K60; 1080p IP livestream
2024–26	V-RAPTOR [X] / XE / XL [X]	8K VV GS 35.4MP	40.96×21.60; 8192×4320	Global shutter + HFR	Multi 12G-SDI

2.3 Sony Cinema Line & Alpha (2008–2026)

Primary sources: Sony Cinema Line chart; VENICE 2 / BURANO / FX product specs; Imaging Edge Webcam (1024×576); Monitor & Control HDMI/UVC help.

Year	Model	Sensor / max record	Key resolutions	Native USB / UVC note	HDMI/SDI live
2008–10	PMW-EX1/EX3, F35	1/2" / Super 35 CCD	1920×1080	—	HD-SDI / HDMI HD
2011	PMW-F3	Super 35 CMOS	1920×1080 (+RAW option)	—	HD-SDI
2012–14	F5 / F55 / F65	S35 4K / 8K F65	4096×2160 class; F65 8K	—	3G/4K SDI ecosystems
2014–17	FS7 / FS5	S35	3840×2160 / 4096×2160	—	3G-SDI + HDMI
2017–	VENICE (6K)	Full-frame 6K	6048×4032 OG class modes	No consumer UVC	12G-SDI + HDMI
2019	FX9	FF 6K oversample → 4K	4096×2160 / 3840×2160	—	12G-SDI ×2
2014–20	α7S / II / III	FF 12MP	4K60 (III)	Imaging Edge Webcam **1024×576**	HDMI clean 4K/HD
2020–22	FX6 / FX3	FF ~10.2MP eff.	4096×2160 / 3840×2160; 120 fps 4K (FX3)	USB Streaming / Imaging Edge 1024×576; M&C UVC path	FX6: 12G-SDI+HDMI; FX3: HDMI 4K
2022	VENICE 2	FF 8.6K / 6K sensor options	**8.6K 3:2**, 8.2K 17:9, 5.8K 6:5	No plug-and-play UVC webcam	Multi 12G-SDI + HDMI
2022–23	FX30 / ZV-E1	APS-C / FF	4K60/120 paths	USB Streaming on supported bodies	HDMI
2023–26	BURANO	FF Exmor RS	**8632×4856** (8.6K)	USB-C data; M&C lists BURANO	HDMI + 12G-SDI
2021–26	α1 / α9 III / α1 II	FF stacked	8K30 (α1); global shutter stills (α9 III)	USB Streaming on current Alphas	HDMI

> **UVC quirk:** Sony Imaging Edge Webcam is fixed at **1024×576** (16:9) and carries no camera audio. Prefer HDMI→Elgato/Magewell for browser GIFs that need ≥720p.

2.4 Canon Cinema EOS & EOS R (2008–2026)

Primary sources: Canon Europe C80/C400 specs; community UVC reports; diiorio.me UVC camera list (cross-checked).

Year	Model	Sensor / record	Notable modes	USB UVC (verified class)	HDMI/SDI
2008–16	5D Mark II/III/IV	FF still CMOS	1080p → 4K (IV)	EOS Webcam Utility ~1024×576	HDMI
2011–15	C300 / C500 / C100	S35	1080 / 4K RAW (C500)	—	HD/3G-SDI

Year	Model	Sensor / record	Notable modes	USB UVC (verified class)	HDMI/SDI
2015–18	C300 Mark II / C700 / C700 FF	S35 / FF	4K / 4.5K class	—	3G/12G-SDI
2019–21	C500 Mark II / C300 III / C70	FF / S35 DGO	5.9K / 4K	—	12G-SDI + HDMI
2020–22	EOS R5 / R5 C / R3	FF 45MP (R5)	8K RAW (R5/R5 C)	R5 C UVC often **~1088×576 / 1024×576 @30**	HDMI; R5 C also SDI via cage/ops
2024	EOS C400 / C80	FF BSI ~26.7MP total (6202×4300)	6K / 6000×3164; S35 crop; Open Gate via C400 firmware (Dec 2025 notes)	C80 UVC listed **1024×576 30p**	12G-SDI + HDMI (6K RAW HDMI on C80)
2024–25	R5 Mark II / R1	FF stacked	8K / high-speed	Native UVC **1080p30** class on current R bodies	HDMI
2025–26	EOS C50	Cinema EOS compact	4K/6K family	UVC listed **1080p60**	HDMI/SDI per SKU

2.5 Blackmagic Design (2012–2026)

Primary sources: blackmagicdesign.com tech specs (Pocket, PYXIS, URSA Cine); Camera 8.6 UVC webcam release notes.

Year	Model	Sensor pixels / size	Record modes (selected)	HDMI/SDI / UVC
2012–13	Cinema Camera 2.5K / Pocket 1080p	2432×1366 class / 1080	2.5K / HD	HDMI HD
2014–15	Production Camera 4K / URSA	4K S35	4000×2160 class	SDI/HDMI
2015–17	URSA Mini 4K / 4.6K	4608×2592 (4.6K)	4.6K / 4K / HD	12G-SDI + HDMI
2017–19	URSA Mini Pro G1/G2	4.6K	BRAW / ProRes	12G-SDI
2018–20	Pocket 4K	MFT 4K	4096×2160	HDMI ≤1080p60; Camera 8.6+ **UVC webcam**
2019–22	Pocket 6K / G2 / Pro	**6144×3456** S35 (23.10×12.99 mm on Pro)	6K OG; 6K 2.4:1; 3.7K 6:5 ana; 4K DCI; UHD	HDMI 1080p60; UVC webcam mode (firmware 8.6+)
2021	URSA Mini Pro 12K	12288×6480 S35 RGBW	12K / 8K / 4K	12G-SDI
2023	Cinema Camera 6K FF	FF 6048×4032 class	Open Gate 3:2	HDMI; UVC on supported firmware
2024	PYXIS 6K / 12K	FF **6048×4032** or **12288×8040**	3:2 OG; 2.4:1; 6:5 ana 9.6K	12G-SDI / HDMI monitoring
2024–26	URSA Cine 12K LF / 17K 65	LF 36×24 mm; 65mm **51×24 mm** up to **17520×8040**	4K–17K in-sensor scaling	12G-SDI; SMPTE 2110 on 100G model

Blackmagic Video Assist USB-C also exposes a standard webcam of the HDMI/SDI input — useful bridge when the camera itself has no UVC.

2.6 Panasonic Lumix / VariCam (2009–2026)

Year	Model	Sensor	Open Gate / ana (verified)	Live out / webcam
2009–24	GH1→GH7	MFT	GH7: **5.8K 4:3** Open Gate 5776×4336 class	HDMI; Lumix Webcam Software historically 1280×960/720
2010–12	AG-AF100	MFT cinema	1080p	HD-SDI/HDMI
2014–18	VariCam 35 / LT / Pure	S35 4K	4096×2160	4K SDI
2017	EVA1	S35 5.7K	5720×3016 class	SDI/HDMI

Year	Model	Sensor	Open Gate / ana (verified)	Live out / webcam
2019	S1H	FF 24MP	Open Gate 3:2 **6K 6024×4016** @24p; 4:3 ana 3520×2640	HDMI 4K; Webcam Software
2021–24	BS1H / S5 II / S5 IIX	FF	S5 II: OG 3:2 **6000×4000** ≤30p; 4:3 ana 3300×2500 ≤50p	HDMI; USB tethering / webcam paths
2025–26	S1R II / GH7 era hybrids	FF / MFT	High-res stills + OG video lineages	HDMI + USB

2.7 Nikon (2008–2026)

Year	Model	Video milestone	USB streaming
2008	D90	First DSLR with movie mode (720p)	—
2012–17	D800 / D850	1080p → 4K (D850)	Webcam Utility ~1024×768 low fps
2018–20	Z6 / Z7	4K UHD	Webcam Utility
2021–23	Z9 / Z8	8K / 4K120	Native **USB Streaming (UVC/UAC)**: 1080p60/30 or 720p60/30 MJPEG + PCM audio
2023–25	Zf / Z6 III / Z50 II	6K/4K hybrids	UVC on supported bodies / Webcam Utility fallback
2025–26	Nikon ZR (RED partnership era)	Cinema-oriented Z mount	Official manual: USB Streaming UVC/UAC (1080/720 class)

2.8 Fujifilm X / GFX (2016–2026)

Year	Model	Sensor / video	UVC / notes
2016–19	X-T2 / X-H1 / X-T3 / X-T4	APS-C; X-T3 4K60	X Webcam app historically low-res; later native UVC on new bodies
2019–21	GFX100 / GFX100S	44×33 mm 102MP	4K video; Premista / large-format stills
2022–23	X-H2S / X-H2 / X-T5	X-H2S stacked; **6.2K Open Gate** internal	Native USB webcam **1080p60** class on current X/GFX
2023–26	GFX100 II / X-M5 / GFX Eterna	GFX100 II 8K video; Eterna cinema color	UVC 1080p60; Premista gate for anamorphic workflows

2.9 Action, gimbal & specialty UVC / capture bridges

Device	Typical record	What browser sees	Notes
GoPro HERO9–13	5.3K / 4K120 / 1080	Webcam mode often 1080p or less via GoPro Webcam	HyperSmooth; digital lenses change FOV/aspect
DJI Osmo Action / Pocket 3 / Ronin 4D	4K–6K / full-frame Ronin 4D	USB webcam or HDMI clean → capture	Pocket 3 1″ sensor; Ronin 4D Zenmuse X9
Insta360 X / Ace	360 / 8K class	Reframed 16:9 webcam exports	Aspect after reframe matters for PR #14 detect
Elgato Cam Link 4K	HDMI in	UVC: 480p/576p/720p/1080i/1080p60 / **2160p30** (60 on latest HW)	Primary bridge for cinema HDMI→browser
Magewell USB Capture HDMI 4K Pro	HDMI in ≤4096×2160 60	High-quality UVC capture	Prefer for DCI 4096×2160
Blackmagic UltraStudio / DeckLink + Video Assist	SDI/HDMI	Assist USB-C webcam of program feed; UltraStudio is Thunderbolt SDK not UVC	Use Assist for zero-driver UVC
Atomos Ninja / Shogun	HDMI/SDI record	Pass-through to capture card	ProRes/RAW record ≠ UVC size

3. Major Smartphones — Year-by-Year Camera Technical Reference (2007–2026)

Photo megapixels are vendor marketing counts; browser/webcam columns emphasize what `getUserMedia` / Continuity Camera / DeviceAsWebcam typically negotiate. Prefer live `getSettings()` over assuming these tables.

3.1 Apple iPhone (2007–2026)

Sources: Apple tech specs (current + archive), Apple developer camera compatibility notes, Continuity Camera WWDC materials, Final Cut Camera 2.0 / Open Gate reporting for 17 Pro.

Year	Model(s)	Rear still	Front	Video record (max class)	Aspects	Browser / Continuity / UVC behavior
2007	iPhone	2 MP	—	Still only	4:3	No video / no getUserMedia era
2008	iPhone 3G	2 MP	—	Still only	4:3	—
2009	iPhone 3GS	3 MP	—	VGA 30	4:3	Early mobile Safari limits
2010	iPhone 4	5 MP	VGA	720p30	4:3 / 16:9 video	—
2011	iPhone 4S	8 MP	VGA	1080p30	4:3 / 16:9	—
2012	iPhone 5	8 MP	1.2 MP	1080p30	16:9 screen era	—
2013	iPhone 5s	8 MP	1.2 MP	1080p30; slo-mo 120	4:3 / 16:9	—
2014	iPhone 6 / 6 Plus	8 MP	1.2 MP	1080p60; slo-mo 240 (Plus)	16:9	—
2015	iPhone 6s / 6s Plus	12 MP	5 MP	**4K30**	4:3 / 16:9	getUserMedia matures on iOS
2016	iPhone 7 / 7 Plus	12 MP (+ tele Plus)	7 MP	4K30	dual-lens Plus	—
2017	iPhone 8 / X	12 MP dual (X)	7 MP TrueDepth (X)	4K60	19.5:9 X display	Portrait sensor vs landscape lock critical
2018	iPhone XS / XR	12 MP	7 MP	4K60	19.5:9 / LCD XR	—
2019	iPhone 11 / 11 Pro	12 MP dual/triple	12 MP	4K60	ultra-wide 120°	Multi-camera labels for PR #14 lens detect
2020	iPhone 12 / 12 Pro	12 MP	12 MP	4K60; ProRes later via update on Pro	20.2:9-class	—
2021	iPhone 13 / 13 Pro	12 MP; Pro cine modes	12 MP	4K60; ProRes; macro	Sensor crop modes	Safari getUserMedia secure-context
2022	iPhone 14 / 14 Pro	Pro: **48 MP** quad-pixel main	12 MP	4K60; Action mode; ProRes	4:3 48MP / 16:9 video	**Continuity Camera** as Mac webcam (typically ≤1920×1080 or 1920×1440 class)
2023	iPhone 15 / 15 Pro	48 MP main; USB-C	12 MP	4K60; Log / ProRes on Pro	USB-C enables external workflows	Continuity + USB capture paths
2024	iPhone 16 / 16 Pro	48 MP; Pro 4K120	12 MP	**4K120** (Pro); 48 MP ultrawide (Pro)	4:3 / 16:9 / cinematic	Same Continuity bounds; high-fps not exposed as UVC 120

Year	Model(s)	Rear still	Front	Video record (max class)	Aspects	Browser / Continuity / UVC behavior
2025	iPhone 17 / 17 Air / 17 Pro	17: Dual Fusion **48 MP** main+UW (Apple specs). Pro family: triple 48 MP + Square/Center Stage front; Open Gate via Final Cut Camera 2.0 on Pro (**4224×3024** 4:3 ProRes RAW class, external SSD)	17: **18 MP Center Stage**	4K Dolby Vision 24–60; front 4K60	4:3 Open Gate; 16:9 deliverable; display ~19.5:9	Safari WebKit getUserMedia; Continuity still HD/1440-class — Open Gate is *recording*, not Continuity UVC
2026 (as of 2026-11-05)	Follow-on / "e" refreshes	As of this document date, treat any 2026 "iPhone 18 / 17e / 16e" claims as **unverified unless Apple publishes specs**. Re-fetch apple.com/iphone/*/specs before coding new EXACT_SIZES entries.				

> **Continuity Camera + gifmakerlive:** On macOS, the iPhone appears as a `videoinput`. Do not assume 4K. PR #13 constraint fallbacks must tolerate OverconstrainedError when asking for 3840×2160. PR #14 `EXACT_SIZES` already maps 1920×1440 → phone-3-4 and 4032×3024 → phone-4-3 for still-ish frames.

3.2 Samsung Galaxy S / Note / Ultra & Z (2010–2026)

Year	Flagship	Rear highlights	Video	Display / fold aspects	Webcam / UVC
2010–12	Galaxy S → S III	5–8 MP	1080p	16:9	—
2013–15	S4–S6 / Note 4–5	16 MP class	4K30 (S6/Note5)	16:9	—
2016–18	S7–S9 / Note 8–9	Dual cams emerge	4K60	18.5:9	—
2019	S10 / Note 10	Triple + ToF	4K60 / HDR10+	19:9	—
2020	**S20 Ultra**	**108 MP** + 100× Space Zoom marketing	**8K24** first Ultra	20:9	Third-party apps
2021–23	S21–S23 Ultra	108→200 MP main	8K30; 4K60	19.3:9–20:9	Dealer apps / DEX
2024	S24 Ultra	200 MP	8K30; 4K120	Near 19.3:9 / 1:1 crop modes	Android 14+ DeviceAsWebcam on supporting builds
2025	S25 Ultra	200 MP main	8K30 on main/UW/5×; 4K120 main/UW	Portrait lock vs fold cover	DeviceAsWebcam where OEM enabled
2026	S26 Ultra	200 MP f/1.4-class main (Samsung marketing)	8K30; APV/pro video paths reported	Same family	Samsung adds native USB webcam on S26 series (2026 reporting)
2019–26	Z Fold / Z Flip	Cover + under-display / punch ratios	4K	Cover ~22:9; inner ~6:5 / 4:3-ish	PR #14 must handle extreme portrait aspects via closestAspect deltas

3.3 Google Pixel (2016–2026)

Year	Model	Rear	Video	DeviceAsWebcam / notes
2016–17	Pixel / Pixel 2	12.3 MP	4K30	—
2018–19	Pixel 3 / 4	12.2 MP (+ tele 4)	4K60	—
2020	Pixel 5	12.2 + UW	4K60	—

Year	Model	Rear	Video	DeviceAsWebcam / notes
2021–22	Pixel 6 / 7 Pro	50 MP main era	4K60; 8K via Video Boost later	—
2023–24	Pixel 8 / 8 Pro	50 MP	4K60; Pro Res zoom	**Android 14 QPR1 DeviceAsWebcam** (Pixel first-party support)
2024–25	Pixel 9 / 9 Pro	50 MP	8K via boost; 4K60 HDR	UVC advertises MJPEG 480/720/1080; YUYV ≤720p30 on USB2; 4K30 MJPEG possible per AOSP guidance
2025–26	Pixel 10 Pro	50 MP main	8K24/30 Video Boost; 4K60 10-bit HDR	Same DeviceAsWebcam stack

> **AOSP DeviceAsWebcam contract (engineering):** Formats MJPEG + YUYV. Vendor guidance: advertise 640×480, 1280×720, 1920×1080 @30. USB2 bandwidth ⇒ uncompressed max ~720p30; MJPEG can claim 4K30. macOS + USB3 cables have a known Apple UVC driver issue (AOSP docs).

3.4 Other major smartphone camera flagships

Line	Years	Camera / sensor notes	Aspect / video	UVC / monitor
Sony Xperia 1 I–VII / Pro-I	2019–2026	1″ on Pro-I; Cinema Pro app	**21:9** display & framing; 4K120 class on later	External monitor / UVC accessory workflows; 21:9 must map via closestAspect (near scope/univisium—may need EXACT_SIZES additions)
Xiaomi 12S/13/14/15/16 Ultra	2022–2026	Leica co-engineered; **1-inch** main on Ultra	8K / 4K60; 4:3 stills	OEM USB webcam varies by HyperOS build
Vivo X100 / X200 Ultra	2023–2026	Zeiss; large tele sensors	4K / 8K class	Region firmware dependent
Huawei Pura / Mate	2023–2026	Ultra lighting / RYYB histories	4K; variable zoom	HarmonyOS USB camera modes

4. Master Resolution, Aspect Ratio & Constraints Matrix

This section is the lookup both PRs should treat as normative. Values marked `EXACT` are already hard-coded in PR #14 `EXACT_SIZES`.

4.1 Exact size → profileId (shipped in PR #14)

Input W×H	profileId	Decimal aspect	Category	Typical source
4096×2160	dci-17-9	1.8963	cinema	DCI 4K
2048×1080	dci-17-9	1.8963	cinema	DCI 2K
4096×1716	scope-239	2.3869	cinema	Scope container
2048×858	scope-239	2.3869	cinema	Scope 2K
3996×2160	flat-185	1.8500	cinema	Flat
1998×1080	flat-185	1.8500	cinema	Flat 2K
3840×1920	univisium-200	2.0000	cinema	Univisium UHD
1920×960	univisium-200	2.0000	cinema	Univisium HD
3840×2160	uhd-16-9	1.7778	cinema/phone	UHD
4096×3072	anamorphic-43	1.3333	cinema	4:3 gate
2880×2160	anamorphic-43	1.3333	cinema	4:3 gate
2048×1536	anamorphic-43	1.3333	cinema	4:3 gate
6000×4000	open-gate-32	1.5000	cinema	Lumix OG / FF still
6048×4032	open-gate-32	1.5000	cinema	BMD PYXIS 6K OG
5760×3840	open-gate-32	1.5000	cinema	OG class
4608×3072	open-gate-32	1.5000	cinema	OG class (near ALEXA 35)
2160×3840	phone-9-16	0.5625	phone	UHD portrait
1080×1920	phone-9-16	0.5625	phone	FHD portrait
720×1280	phone-9-16	0.5625	phone	HD portrait
1080×1440	phone-3-4	0.7500	phone	Continuity 4:3 portrait-ish
3024×4032	phone-3-4	0.7500	phone	12MP portrait still
1080×1080	phone-1-1	1.0000	phone	Square
2160×2160	phone-1-1	1.0000	phone	Square UHD
4032×3024	phone-4-3	1.3333	phone	12MP landscape still

Input W×H	profileId	Decimal aspect	Category	Typical source
4000×3000	phone-4-3	1.3333	phone	12MP landscape

> **Recommended EXACT_SIZES additions before merge** (verified hardware, not yet in PR #14): 4608×3164 → open-gate-32 (ALEXA 35 OG), 4448×3096 → open-gate-32 (ALEXA LF), 6144×3240 → uhd-16-9 or custom DCI-ish (KOMODO; aspect 1.896 ≈ DCI), 6144×3456 → open-gate-32 (BMD Pocket 6K), 8192×4320 → uhd-16-9 (V-RAPTOR FF), 4224×3024 → phone-4-3 (iPhone 17 Pro Open Gate), 1024×576 → uhd-16-9 (Sony/Canon webcam utility), 1920×1440 → phone-4-3 (Continuity Camera 4:3). KOMODO 6144×3240 is exactly 17:9 / DCI aspect — map to dci-17-9, not 16:9.

4.2 GIF output height at standard width tiers (aspect-preserving match)

Formula: `height = round(width / aspect)`, then reject if `width*height > 1_600_000` or `height > 4096`. Desqueeze profiles use `aspect_out = aspect_source × factor`.

profileId	Aspect	100	160	240	320	480	640	800
phone-9-16	0.5625	178	284	427	569	853	1138	1422
phone-16-9	1.7778	56	90	135	180	270	360	450
phone-4-3	1.3333	75	120	180	240	360	480	600
phone-3-4	0.7500	133	213	320	427	640	853	1067
phone-1-1	1.0000	100	160	240	320	480	640	800
dci-17-9	1.8963	53	84	127	169	253	338	422
uhd-16-9	1.7778	56	90	135	180	270	360	450
scope-239	2.3900	42	67	100	134	201	268	335
univisium-200	2.0000	50	80	120	160	240	320	400
flat-185	1.8500	54	86	130	173	259	346	432
open-gate-32	1.5000	67	107	160	213	320	427	533
anamorphic-43	1.3333	75	120	180	240	360	480	600

4.3 Anamorphic desqueeze worked examples

Squeezed source	Factor	Effective aspect	GIF @320w	GIF @640w	Use
2880×2160	×1.33	1.7733	320×180	640×361	PR #14 shipped
2880×2160	×1.5	2.0000	320×160	640×320	PR #14 shipped
2880×2160	×1.8	2.4000	320×133	640×267	PR #14 shipped
2880×2160	×2	2.6667	320×120	640×240	PR #14 shipped
3328×2790	×2	2.3857	320×134	640×268	ALEXA 35 6:5 + 2× glass

Squeezed source	Factor	Effective aspect	GIF @320w	GIF @640w	Use
4096×3072	×2	2.6667	320×120	640×240	4:3 gate + 2×
1920×1080	×1.33	2.3644	320×135	640×271	PR #14 shipped
2880×2160	×1.25	1.6667	320×192	640×384	**Recommended add** (not yet in PR #14)
2880×2160	×1.3	1.7333	320×185	640×369	**Recommended add** (not yet in PR #14)
2880×2160	×1.6	2.1333	320×150	640×300	**Recommended add** (not yet in PR #14)

4.4 UVC / getUserMedia fallback quirk matrix

Source class	Common advertised modes	PR #13 guidance	PR #14 guidance
Built-in laptop webcam	640×480, 1280×720, 1920×1080	DEFAULT 1280×720 ideal; fall to `video:true`	uhd-16-9 / match
Sony/Canon legacy webcam utility	**1024×576**	Do not require 1280 height; Overconstrained common	closestAspect → uhd-16-9
Nikon native UVC	1920×1080 / 1280×720 @30/60 MJPEG	Prefer exact deviceId after permission	uhd-16-9
Fujifilm / modern Canon R UVC	1080p30–60	Same	uhd-16-9
BMD Pocket UVC / Video Assist	~1080p60 class	Label often contains Blackmagic → cinema class	CINEMA_LABEL regex hit
Elgato Cam Link 4K	720–2160p (30; 60 on new HW)	Enumerate after plug (`devicechange`)	Exact 3840×2160 / 1920×1080
Magewell 4K HDMI	up to 4096×2160	May expose DCI; try 4096 mode in CAMERA_MODES	dci-17-9 exact
Continuity Camera (iPhone)	≤1080p16:9 or 1920×1440 4:3	Facing controls only when deviceId empty	phone-* profiles; orientation lock
Android DeviceAsWebcam	480/720/1080 MJPEG; maybe 4K MJPEG	Hot-plug USB; re-enumerate	match + phone class via label
ARRI/RED SDI via capture	Usually 1080p or UHD monitor	Never assume Open Gate pixels	Detect cinema from label; allow manual profile

4.5 Orientation & live constraint switching

- `orientSource`: lock `portrait|landscape` may flip W/H without reopening the camera.
- `cameraModes(capabilities)` filters `CAMERA_MODES` against `getCapabilities().width/height`.
- `constraintsForMode(w,h)` sets ideal width, height, and aspectRatio — then PR #13-style fallbacks should still apply if the track rejects.
- Listen to `resize`, and where available `requestVideoFrameCallback`, plus `screen.orientation` changes to refresh `chooseLiveSize`.
- Framing: `crop` fills output (may trim source); `fit` letterboxes; desqueeze always crops after horizontal unsqueeze in `placeSource`.

4.6 MediaRecorder codec matrix (PR #13)

Order	MIME	Container ext	Notes
1	`video/webm;codecs=vp9`	webm	Preferred Chromium
2	`video/webm;codecs=vp8`	webm	Fallback
3	`video/webm`	webm	Browser default webm

Order	MIME	Container ext	Notes
4	`video/mp4;codecs=avc1.42E01E`	mp4	Baseline AVC
5	`video/mp4;codecs=avc1`	mp4	Generic AVC
6	`video/mp4`	mp4	Safari-friendly last resort
7	If all fail → direct canvas grab @ normalizeFps(1–30, default 10) for DIRECT_GIF_SECONDS (5s), enforcing FRAME_PIXEL_BUDGET		

5. Official Technical Documentation & Standards Bibliography

Hyperlinks are clickable in this PDF. Prefer manufacturer pages over secondary blogs when updating numbers.

5.1 Web platform & USB standards

- W3C Media Capture and Streams — https://www.w3.org/TR/mediacapture-streams/
- W3C Screen Orientation API — https://w3c.github.io/screen-orientation/ · MDN: MDN Screen Orientation
- USB-IF Video Class (UVC) 1.5 document set — https://usb.org/document-library/video-class-v15-document-set
- AOSP DeviceAsWebcam — https://source.android.com/docs/core/camera/webcam
- Apple Continuity Camera (WWDC22) — https://developer.apple.com/videos/play/wwdc2022/10018/

5.2 Cinema & camera manufacturers

- ARRI ALEV Sensors — arri.com …/alev-sensors
- ARRI ALEXA Mini LF FAQ — Mini LF FAQ
- ARRI ALEXA 35 — ALEXA 35
- RED Compare Specs — reddigitalcinema.com/compare-specs
- RED MONSTRO 8K VV Tech Specs — docs.red.com MONSTRO
- Blackmagic Pocket Tech Specs — Pocket techspecs
- Blackmagic PYXIS — PYXIS
- Blackmagic URSA Cine — URSA Cine
- Sony Monitor & Control HDMI/UVC — helpguide.sony.net … connecting_hdmi_uvc
- Sony Imaging Edge Webcam — support.d-imaging.sony.co.jp/app/webcam
- Canon EOS C80 Specs — canon-europe.com …/eos-c80/specifications
- Canon EOS C400 Specs — canon-europe.com …/eos-c400/specifications
- Panasonic GH7 Specs — panasonic.com …/dc-gh7.specs
- Nikon Z9 USB Streaming (UVC/UAC) — onlinemanual.nikonimglib.com/z9 … UVC
- Nikon ZR Live streaming UVC/UAC — onlinemanual.nikonimglib.com/zr
- Fujifilm X-H2S — fujifilm-x.com …/x-h2s

- Elgato Cam Link 4K Specs — help.elgato.com Cam Link 4K
- Magewell USB Capture HDMI 4K Pro — magewell.com tech-specs

5.3 Smartphones

- Apple iPhone 17 Tech Specs — apple.com/iphone-17/specs (regional mirrors such as /kw/)
- Apple iPhone compare — apple.com/iphone/compare
- Apple Final Cut Camera 2.0 (Open Gate announcement) — apple.com/newsroom/2025/09/...
- Google Pixel webcam support — support.google.com/pixelcamera/answer/14274129
- Samsung Galaxy S26 Ultra — samsung.com .../galaxy-s26-ultra

5.4 Repository PRs this document specifies

- PR #13 — Open and record any camera the browser can see — github.com/qramarq/gifmakerlive/pull/13 · primary code `web/camera.js`
- PR #14 — Size GIF output from the active camera frame — github.com/qramarq/gifmakerlive/pull/14 · primary code `web/camera-format.js`, `crates/gif-core/src/camera.rs`, `crates/studio/src/lib.rs`

Appendix A — Merge checklist for PR #13 + PR #14

1. Keep stream open/close exclusively in `camera.js`; format math exclusively in `camera-format.js` / Rust `camera.rs`.
2. Wire `watchLiveVideo` → `chooseLiveSize` → `detectCameraFormat` → `outputDimensions`/`liveMatchSize`.
3. Add missing desqueeze factors 1.25 / 1.3 / 1.6 and EXACT_SIZES rows listed in §4.1 warn box.
4. Extend `CAMERA_MODES` only with sizes that appear in real `getCapabilities()` across Continuity, DeviceAsWebcam, Cam Link, and native UVC (avoid 8K ideals that only exist as files).
5. Document in README that Open Gate / 8K cinema rasters are for file import + profile selection; live UVC is usually HD/UHD.
6. Reject GIF plans when `tooLarge` / `assertLoopBudget` fails — never silently downsample without UI notice.

Document generated for Marq Williams as the unified engineering reference behind gifmakerlive PR #13 and PR #14. All numeric claims were checked against manufacturer or standards documentation available on 2026-11-05 (America/New_York). Re-verify before encoding any new EXACT_SIZES entry.